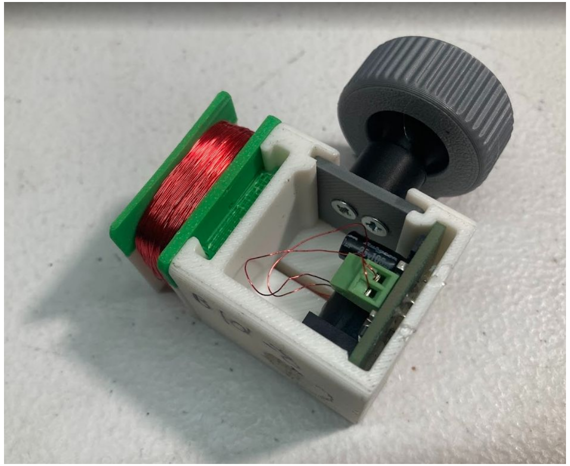


free energy

contactless bicycle power dynamo (mag'neat'o)

William Kennedy (freakylamps.com)
Open Hardware Summit 2026
Berlin 5.25.2026
https://github.com/zumdar/ContactlessBikeDynamo

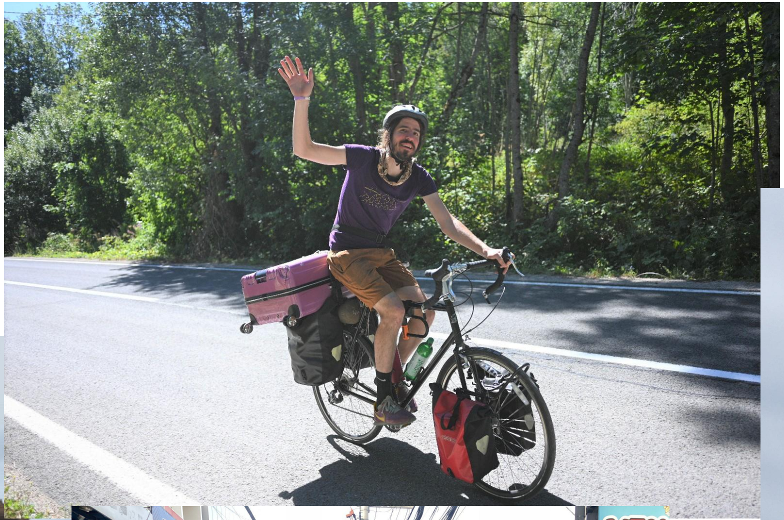
Overview

- Who am I
- Demo
- Dynamo tech and history
- Eddy Currents and Physics
- Make a dynamo
- TEST
- Hang out

Bike Demo

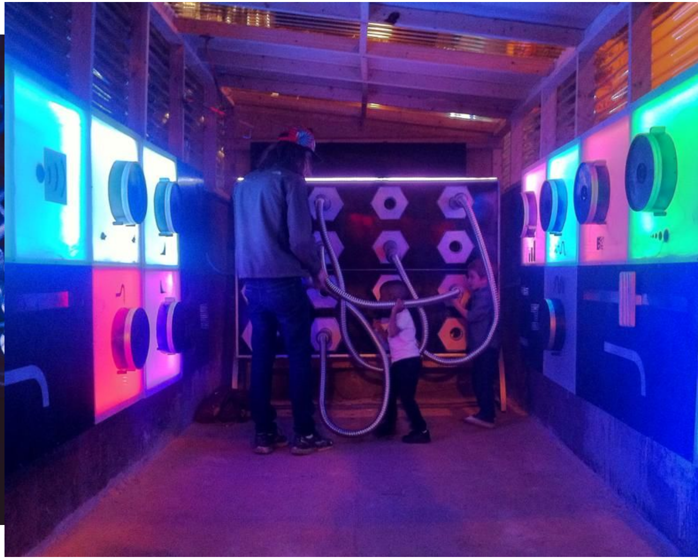

Made by JM

Who am I







What is a bicycle DYNAMO?

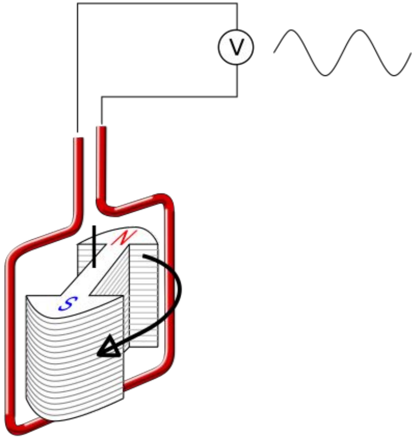
- small device that converts mechanical energy into electric energy to power your bike’s lights while you ride.

Hub Dynamo

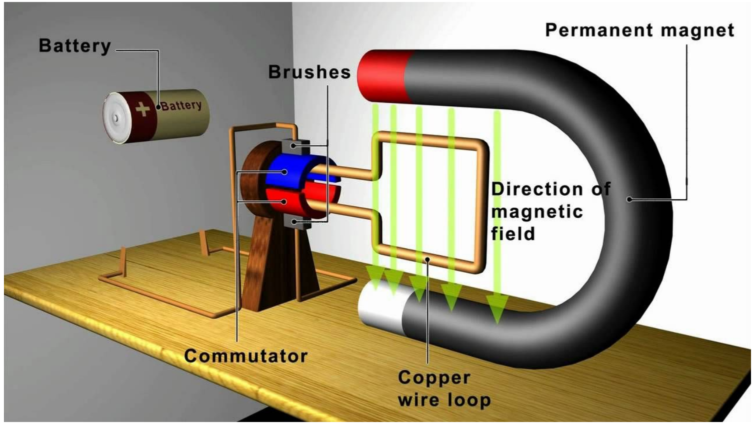
Bottle Dynamo

Dynamo? Generator? Magneto? Alternator?

- Generator
  - broad term usually meaning a machine that generates **AC**
- Dynamo
  - uses an electromagnet and a commutator with spinning wire to create **DC**
- Magneto
  - uses a permanent magnet and spinning wire to create **AC** voltage - typically pulsed, not smooth
- Alternator
  - uses a field electromagnet (the rotor) which spins inside of a of stationary coils of wire (the stator) - creates **AC**



Alternator



Dynamo

Hub Dynamos

- History
    - invented in 1936 by Sturmey-Archer in England.
    - high drag and low output energy. Declined in 1980s.
    - Modern high efficiency dynamo engineered by German engineer Wilfried Schmidt in 1995 (Schmidt Original Nabendynamo (SON))
- Advantages
    - High Efficiency and output (3 watts!)
    - very low drag
    - reliable
    - no adjusting , clean cable wiring, harder to steal
- Disadvantages
    - expensive to purchase and do a wheel build around it
    - tied to that one wheel

Bottle/Sidewall/Rim Dynamo

- History
    - 1923: German Robert Bosch invented the modern bicycle dynamo - Post WW1 depression
    - Pre - kerosene lamps, arc lamps with big batteries
    - 1930s - 1960s golden age for bottle dynamos
- Advantages
    - Easy to retrofit
    - lower cost
    - can be disengaged easily
- Disadvantages
    - higher drag and friction
    - more wear and tear (touching wheel)
    - slippage when wet
    - can get knocked out of alignment

Battery Lights

- History
  - 1896–1900, small torch-style lamps using zinc-carbon dry cells were strapped to handlebars or forks.
  - Low self-discharge NiMH battery (LSD-NiMH) in 2005 started allowing better more powerful lights
  - Li-ion and LiPo batteries around 2010 supercharged battery powered bike lights
- Advantage
  - High Power
  - easy to remove
  - rechargeable
- Disadvantages
  - need to recharge
  - easily stolen
  - charging cables…

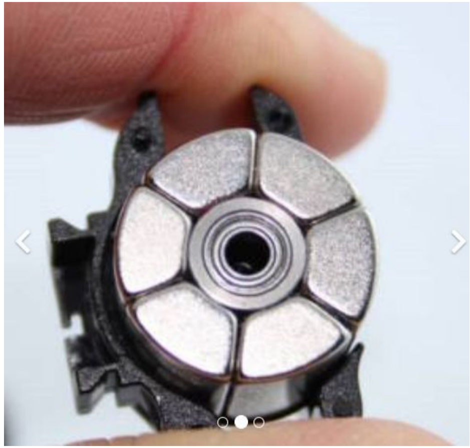
Eddy Current Dynamos

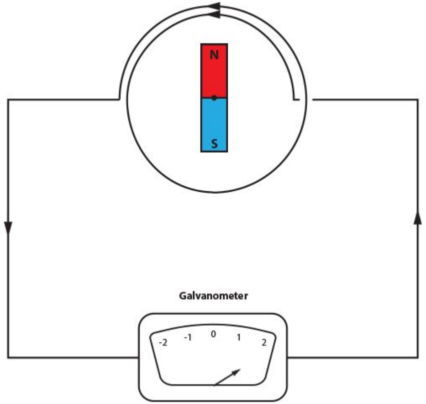
- History
    - Magnic Light (Germany, ~2012)
    - Reelight - Danish
    - Wheelswing - South Korea 2017
- Advantages
    - Zero drag - no friction contact
    - No tire wear
    - Works in wet conditions
    - Easy to install, no wheel rebuild required (unlike hub dynamos)
    - No batteries to replace
    - Magic
- Disadvantages
    - Low power output
    - need to run cables to light

Eddy Current Dynamos

EDDY CURRENTS DYNAMOS!

- Ring of alternating polarity magnets around a bearing
- This ring is placed inside a coil of wire, when the magnets are spun, there is an electric current induced in the wire
- BUT… How does this magnet ring spin?



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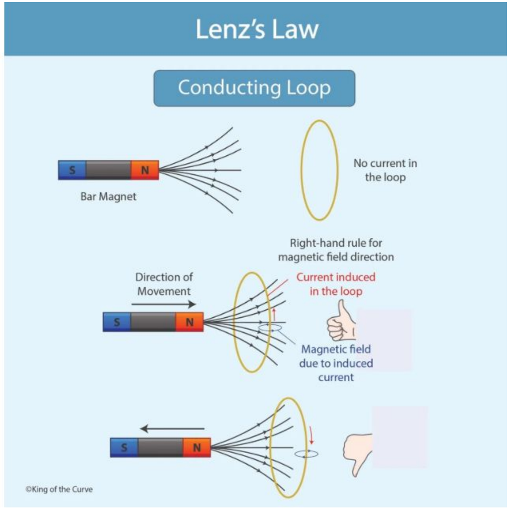
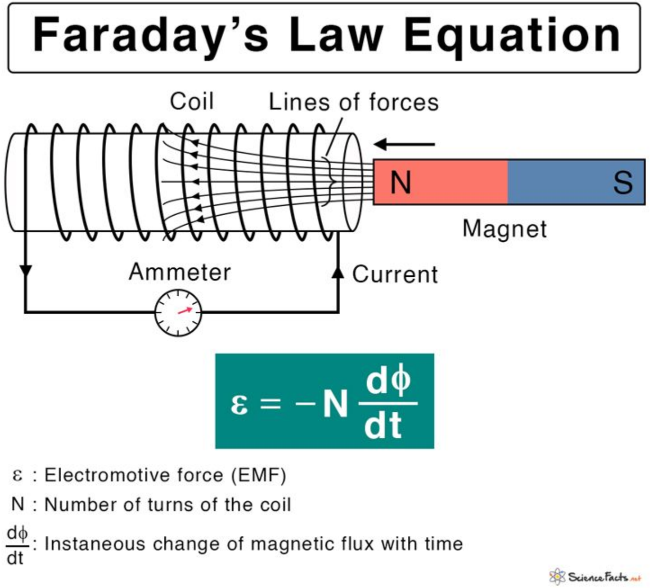
Physics - Electromagnetism

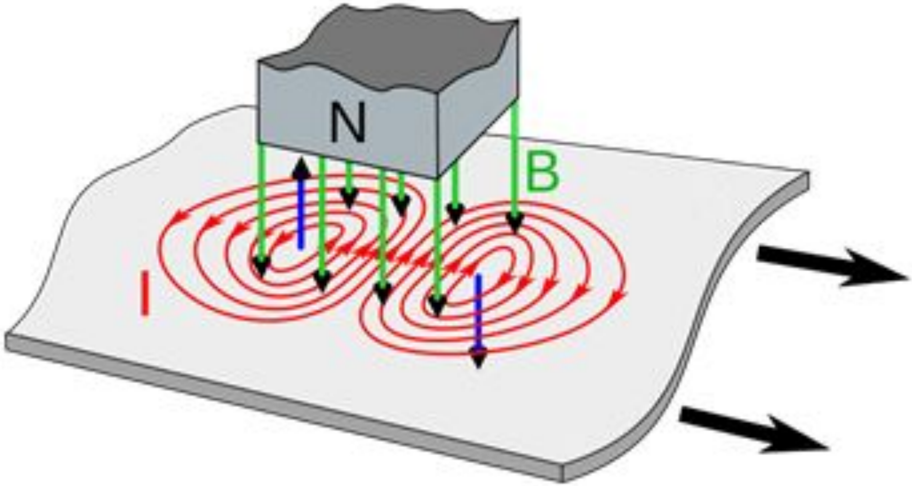
- Magnetic field
  - An invisible region of force surrounding magnets , flowing from north to south
- Faraday's Law
  - Changing magnetic field → induces voltage in a conductor.
- Lenz's law
  - Induced current opposes the change that caused it.

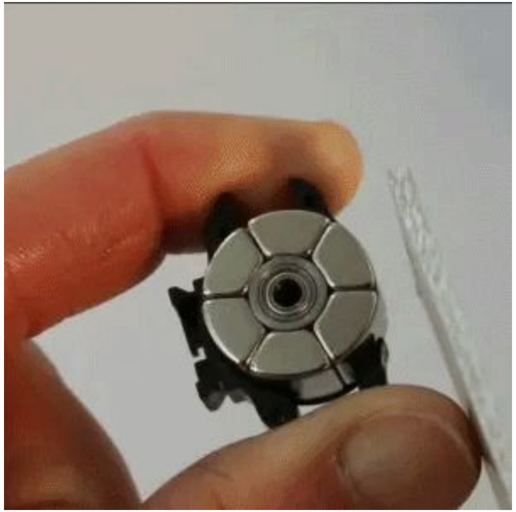




Eddy Currents

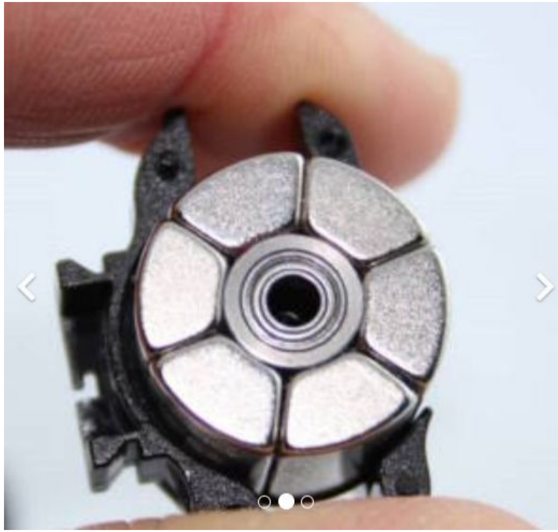
- moving magnet near solid conductor
- The changing field induces voltage throughout the bulk metal (Faraday).
- The resulting currents circulate in loops that resist the magnet's motion (Lenz). Those circulating loops are eddy currents.
- Those currents have their own opposing magnetic field (Lenz)



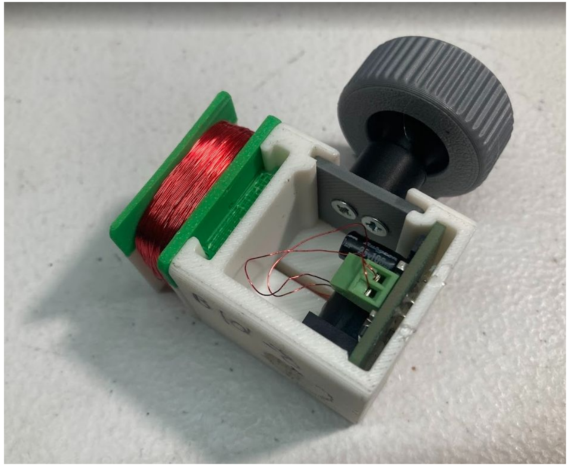


EDDY CURRENTS DYNAMOS!

- Work by NOT touching the moving bicycle rim
- Ring of alternating polarity magnets around a bearing
- This is placed near the aluminum rim of bike
  - Aluminum is electrically **conductive** but not **magnetic**!
- As the wheel spins, the magnets induce an eddy current in the rim of the wheel
- That eddy current has an opposing magnetic field that pushes the magnet ring
- The spinning of the magnet ring in a coil of wire induces a current
- That current is rectified (AC / DC) and sent to bike light



The Open Source Eddy Current Dynamo Rev. 2!



Work to do - Want to get involved?

- get more power out of it?
- circuit to keep light on when you stop pedalling
    - flash or solid

WORKSHOP THYME

Special Thanks

Emma Meyers & Chelsea Dunn - workshop and slide critical feedback

Kevin Bower - mechanical engineering design help with V2

Lucas Ray - mechanical engineering design help with V1

YJ Rodrigues - electrical engineering on V1

Clara Caspard - mechanical engineering design help with V1

References

Patents

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- Eddy Current Generator for Bicycle - 20140070675

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- Magnic Light - Eddy Current explanation
- https://www.sheldonbrown.com/marty_light_hist.html
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